

Soumyadyuti Ghosh

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Education & Research

■ **New York University Abu Dhabi (NYUAD), Abu Dhabi, UAE** 2025 – Present
Postdoctoral Associate, Center for Cyber Security (CCS)
Supervisor: Prof. Michail (Mihalis) Maniatakos

Primary area of work: i) Privacy-preserving computation using Fully Homomorphic Encryption (FHE), with emphasis on encrypted keyword search and abuse detection in end-to-end encrypted messaging, and high-accuracy encrypted arithmetic for smart grids; ii) Class prior-free Positive-unlabeled learning, and practical validation of learning models on real-world fine-grained datasets.

■ **Indian Institute of Technology Kharagpur, India** 2019 – 2025
Ph.D. in Computer Science and Engineering
Supervisors: Prof. Debdeep Mukhopadhyay and Prof. Soumyajit Dey
Secured Embedded Architecture Laboratory (SEAL) and High Performance Real-Time Computing Laboratory (HiPRC)

Area of work: Privacy-preserving computation in Cyber-Physical Systems (CPS), primarily in the Smart Grid domain. Research focuses on identifying vulnerabilities and developing methodologies for secure communication and privacy-preserving computations, while ensuring system stability through cryptographic and information-theoretic techniques such as Differential Privacy, Secure Multi-Party Computation, and Fully Homomorphic Encryption.

■ **Jadavpur University, West Bengal, India** 2012 – 2016
B.E. in Computer Science and Engineering
Graduated with 7.83 CGPA

- Secured 364th rank out of 150,000+ students in **Joint Entrance Examination**, 2012.
- **Higher Secondary:** 86.2% overall and 94.34% in science category, W.B.C.H.S.E, India, 2012.
- **Secondary:** 82% overall and 96.34% in science category, W.B.B.S.E, India, 2010.

Work Experience

■ **Wipro Digital, Bangalore, India** 2016 – 2019
Senior Developer and Project Lead

- Designed and developed a PaaS model with a secure end-to-end communication gateway for medical devices, supporting device registration, data synchronization, remote access, file transfer, network bandwidth throttling, concurrency management, priority handling, and logging.
- Developed an automated robot capable of identifying random objects in industrial environments and performing essential operations using the Microsoft Azure platform.
- Integrated soil moisture sensors to determine critical moisture levels and maintained real-time time-series data on Bluemix and GE Predix services.

Research & Internship Experience

■ **Visiting Research Scholar, Center for Cyber Security (CCS), New York University Abu Dhabi (NYUAD), 2023**

Supervisor: Prof. Michail (Mihalis) Maniatakos

- Researched accurate floating-point division in the CKKS homomorphic encryption scheme, applied to computations on encrypted Smart Grid data, enabling secure and precise grid functionalities while preserving confidentiality.

■ **Summer Internship, Department of Computer Science and Engineering, Indian Institute of Technology (IIT) Kharagpur, 2015**

Supervisor: Prof. Chittaranjan Mandal

- Designed an automated problem generation and solution verification system with randomized, easily drawable instances of various data structures, producing compact questions and automatic feedback based on solution correctness.

■ **Project Internship, Department of Computer Science and Engineering, Jadavpur University, Kolkata, 2016**

Supervisor: Prof. Chandan Mazumdar

- Developed an application to regularly collect useful newspaper data from multiple websites for simplified large-scale data processing using MapReduce on clusters.

Selected Research Projects & Tools

■ **Cyber Security Research in Cyber-Physical Systems**

Sponsored by Haldia Petrochemicals Ltd. and TCG Foundation

- **Smart-Grid Security:** Set up a Hardware-in-the-Loop testbed with OPAL-RT simulator, CompactRIO controller, PMUs, PDC, V/I Power Amplifier, GPS Clock, and automation controller to study the impact of various attacks on power system control loops and propose suitable countermeasures, including secure communication protocols in the smart grid environment.

■ **Next-Generation Secured Internet of Things (IoT)**

Funded by the Department of Science and Technology (DST), Government of India

- **Smart Meter Design:** Developed a low-cost, secure smart meter on Raspberry Pi 4 for integration into a smart grid testbed. Embedded a spoofing-resistant PUF-based authenticated key exchange (AKE) protocol. Simulated environmental changes using a CME bench-top chamber to assess SRAM-PUF reliability and evaluate AKE protocol performance.

■ **Centre on Hardware Security**

Data Security Council of India and Ministry of Electronics and Information Technology, Government of India

- **Assessment of PLCs used in Smart Grids:** Used CompactRIO as PLC and OPAL-RT as plant simulator to realize closed-loop grid functionalities and study vulnerabilities of PLCs under various attack scenarios (e.g., HARVEY, NDSS'17), providing insights into operational challenges in realistic Smart Grid environments.

■ **KAVACH: Key Analysis and Vulnerability Assessment for Cryptographic Hardware**

Co-developed with PROC SHIELD Private Limited

- Co-developed KAVACH, an integrated GUI-based platform for side-channel and fault analysis of cryptographic hardware. It controls FPGAs and oscilloscopes, captures power and electromagnetic traces from cryptographic co-processors, and integrates ChipWhisperer Lite for low-cost trace acquisition and fault injection. KAVACH supports Correlation Power Analysis (CPA) for AES-128 key recovery and clock-glitch attacks that generate faulty ciphertexts for Differential Fault Analysis (DFA). PROC SHIELD is incubated under the Information Security Education and Awareness (ISEA) initiative of the Ministry of Electronics and Information Technology (MeitY) and the Centre on Hardware Entrepreneurship Research

& Development (HERD).

Publications

■ Newly Accepted Publications

- “HARP: Privacy-Preserving High-Accuracy Real-Time Pricing with Precise Encrypted Inverse for Smart Grids.” In **European Symposium on Research in Computer Security (ESORICS)**, 2026.
- “A Learning-Enabled Framework for Profit-Driven Attack Synthesis in Electricity Markets.” In **International Conference on Security, Privacy and Applied Cryptographic Engineering (SPACE)**, 2026.
- “PULSE: Support-Calibrated Class Prior-Free Positive-Unlabeled Learning with Hard-Negative Ranking and Lower-Tail Coverage.” In **ACM International Conference on Information and Knowledge Management (CIKM)**, 2026.

■ Published Papers

- “Demand Manipulation Attack Resilient Privacy Aware Smart Grid Using PUFs and Blockchain.” In **Applied Cryptography and Network Security Workshops (ACNS)**, 2021.
- “Is the whole lesser than its parts? Breaking an aggregation based privacy aware metering algorithm.” In **25th Euromicro Conference on Digital System Design (DSD)**, 2022.
- “SMarT: A SMT Based Privacy-Preserving Smart Meter Streaming Methodology.” In **International Conference on Security, Privacy and Applied Cryptographic Engineering (SPACE)**, 2022.
- “hello? is there anybody in there? Leakage assessment of differential privacy mechanisms in smart metering infrastructure.” In **International Conference on Applied Cryptography and Network Security (ACNS)**, Springer, 2024.
- “Differentially Private Real Time Pricing Control for Smart Grids.” **ACM Transactions on Cyber-Physical Systems (TCPS)**, 2025.
- “Pay What You Spend! Privacy-Aware Real-Time Pricing with High Precision IEEE 754 Floating Point Division.” In **ACM Asia Conference on Computer and Communications Security (AsiaCCS)**, 2025. [Selected for contributed talk in TPMPC]

Selected FHE Research

- **HARP: Privacy-Preserving High-Accuracy Real-Time Pricing with Precise Encrypted Inverse for Smart Grids.** HARP develops a privacy-preserving real-time pricing protocol for smart grids using approximate leveled fully homomorphic encryption. Its central technical challenge is computing a precise inverse over encrypted floating-point data, since numerical errors can be amplified by the pricing feedback loop and disturb generation scheduling. HARP addresses this through an optimized Taylor-series approximation designed to control FHE noise while retaining high precision. The resulting encrypted inverse supports pricing computations without exposing consumers’ fine-grained electricity-consumption profiles. In realistic experiments, HARP closely mirrors the original plaintext real-time-pricing computation, with a median generation-scheduling error below 10^{-5} and a maximum below 10^{-4} .
- **Privacy-preserving security analysis for encrypted messaging.** Ongoing work explores FHE-based methods for authorized search and coordinated-abuse detection in end-to-end encrypted messaging while limiting what the service learns about messages, queries, and low-support reports.

Professional & Academic Service

- Talks, Tutorials, and Seminars:

- Talk at the Workshop on Architecture for Security and Privacy (WASP), Indian Institute of Technology Kharagpur, 2024.
- Tutorial lecture on Secure Multi-Party Computation (MPC) and Fully Homomorphic Encryption (FHE) in the Privacy Enhancing Technologies (PETs) course at Indian Institute of Technology Kharagpur, 2025.
- Invited guest lecture at Boot Camp, Indian Institute of Technology Kanpur, 2025.
- Invited contributed talk at the Workshop on Theory and Practice of Multi-Party Computation (TPMPC), 2025.
- Seminar talk as part of the CCSAD Cybersecurity Seminar Series, New York University Abu Dhabi (NYUAD), 2025.
- **Teaching Assistant (Indian Institute of Technology Kharagpur):** Cryptography and Network Security, Privacy Enhancing Technologies (PETs), Computational Foundations of Cyber-Physical Systems, High-Performance Computer Architecture, Programming and Data Structures (theory and lab), Computer Organization Lab, Formal Language and Automata Theory, Hardware Security.
- **Journal and Proceedings Reviewer:** IEEE TCAD, IEEE TSG, IEEE TIFS, IEEE TDSC, ACM TECS, ACM TCPS, IACR TCHES, IEEE PESL.
- **Conference and Workshop Reviewer:** USENIX Security Symposium; Network and Distributed System Security Symposium (NDSS); Cryptographic Hardware and Embedded Systems (CHES); IEEE/ACM International Conference on Computer-Aided Design (ICCAD); Design, Automation & Test in Europe Conference & Exhibition (DATE); Design Automation Conference (DAC); ACM Asia Conference on Computer and Communications Security (AsiaCCS); European Symposium on Research in Computer Security (ESORICS); Applied Cryptography and Network Security (ACNS); Euromicro Conference on Digital System Design (DSD); IEEE International Symposium on Hardware Oriented Security and Trust (HOST); International Conference on Security, Privacy and Applied Cryptographic Engineering (SPACE); Workshop on Attacks and Solutions in Hardware Security (ASHES); and Asian Hardware Oriented Security and Trust Symposium (AsianHOST).
- **Event Organization and Service:**
 - Contributed to the organization of the Workshop on Cyber-Physical System Security, Indian Institute of Technology Kharagpur, India, 2019.
 - Contributed to the organization of the International Conference on Security, Privacy and Applied Cryptographic Engineering (SPACE), Indian Institute of Technology Kharagpur, India, 2021.
 - Contributed to workshop on Architecture for Security and Privacy (WASP), Indian Institute of Technology Kharagpur, 2024.
 - Contributed to workshop on Machine Learning and Hardware Security, IIT Kharagpur, 2025.
 - Supported the Middle East and North Africa (MENA) regional activities of Cybersecurity Awareness Worldwide (CSAW), organized by the Center for Cyber Security at NYUAD, including serving as a judge and evaluator for the BioHack 3D competition.

Skills

Programming: C, C++, Java, Python, MEAN and MERN stack, MapReduce.

Technical:

- **Differential Privacy:** Google's differential privacy libraries.
- **Secure Multi-Party Computation:** SPDZ, MP-SPDZ, SCALE-MAMBA, SecFloat, MOTION, Blaze, Swift, EzPC, Falcon, Bristol Fashion MPC Circuits, CBMC-GC.
- **Homomorphic Encryption:** Microsoft SEAL, node-seal, OpenFHE, HEAAN, and Pyfhel.

- **Foundations of Cyber-Physical Systems (CPS):**
 - Stochastic modeling and simulation of CPS.
 - Real-time pricing, stability, and privacy-preserving computation in Smart Grids.
 - Matlab, Simulink, GridLAB-D, optimization tools (e.g., YALMIP).
 - Multi-dimensional Kalman filtering.
- Side-channel analysis of standard cryptosystems.
- Trace acquisition using electromagnetic probes.
- Satisfiability Modulo Theories: Z3 solver.
- **Machine Learning:** Class prior-free Positive-unlabeled learning, PyTorch and TensorFlow.
- Floating-Point Arithmetic: IEEE-754, POSIT.

Academic Courses: Cryptography and Network Security, Computational Foundations of Cyber-Physical Systems, Data Structure and Algorithms, Computer Architecture, Digital Logic, Operating Systems, Computer Networks, Formal Methods, Automata Theory, Database Management, Advanced Machine learning.

Cloud Computing: Microsoft Azure, Casablanca (Azure REST SDK), Amazon Web Services, IBM Bluemix, GE Predix.

Hardware Expertise: OPAL-RT real-time simulator; National Instruments CompactRIO Real-Time Controller (CRIO 9076); Phasor Measurement Units (SEL 421, SEL 2240); Phasor Data Concentrator (SEL-3555); V/I Power Amplifier; GPS Clock (SEL 2404); SEL SynchroWAVE Historian; Power Meters (PM 5320, PM 5330); Raspberry Pi 4 Model B; ChipWhisperer Lite.

Certificates, Awards & Recognitions

- Second place in the “Embedded Security Challenge” at Cyber Security Awareness Week (CSAW), 2020 and 2021.
- Topper in Ph.D. comprehensive examination, IIT Kharagpur.
- Recipient of Wipro Digital “Wunderkind” award, 2018.
- IBM Certified Application Developer – Cloud Platform V1.
- Secured state rank 4 in Mathematics Olympiad, 2008.

Languages

- **Bengali:** Mother tongue
- **English:** Fluent
- **Hindi:** Basic proficiency

Extracurricular Activities

- Captain of the department football team and member of the college football team at Jadavpur University.
- Captain of the department football team at IIT Kharagpur; winner of the Research Scholar Football Championship in 2022 and 2024.
- Runner-up in college Bridge tournament (2015) and winner (2016); regular and competitive Auction and Duplicate Bridge player.

- Regular chess player on Chess.com with a rapid rating of 2267.

References

- **Prof. Debdeep Mukhopadhyay**
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- **Prof. Soumyajit Dey**
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- **Dr. Ajith Suresh**
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I hereby declare that the above particulars are true and correct to the best of my knowledge.

Date: August 31, 2026

Soumyadyuti Ghosh